DIGITAL COMPOSITION SYSTEMS: USER EXPERIENCE STUDY

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ABSTRACT

Digital composition systems shape the way a composer creates a composition. Several novel methods of digital composition have demonstrated that open-ended possibilities in the design of composition interfaces can be expanded on. Generative non-linear processes have demonstrated their usefulness and ability to aid creative processes. This study investigates what tasks are effectively achieved and what problems arise when using two different types of digital composition systems, specifically focusing on linear and non-linear systems.

Keywords

Digital Composition Systems, User Study, Generative Composition

1. COMPOSITION SYSTEMS

Avid Protools and systems similar to it such as Apple Logic and Cubase are considered industry standard digital audio workstations (DAW's) amongst both amateur and professional music producers. Within this group of programs there is little deviation from what would be considered standard user interfaces. Design of these interfaces is closely abstracted from multitrack recorders and mixing consoles historically used in the analogue process of recording to magnetic tape (Theberge 2004). DAW's also have basis in western musical notation, allowing a user to listen to a linear series of sequenced events measured against a grid which represents time and musical measure. This is coupled with the literal abstraction of the piano into the piano roll, with both pitch and velocity being represented as linear sequences of events.

Development of novel composition systems has taken on a new form in recent years as platforms such as MaxMSP have now developed simple user friendly ways to integrate with industry standard DAW's. As designing tools for creating music has become easier, the demand for diversity in digital musical tools for both live music performance and composition has risen. Large online communities now collaborate to design musical tools. MaxMSP and Pure Data have large databases of patches available online with new and novel tools constantly being developed and refined. Popularity of high-level programming platforms for creative purposes has culminated in a collaboration between Cycling74 and Ableton. Max for Live (M4L) is the integration of MaxMSP into Ableton Live as a plug-in. This enables designers to create and distribute MaxMSP patches easily to an even broader audience and allows novice Ableton Live users to integrate experimental composition tools into their workflow easily.

Nodal [9] is an example of a novel composition tool that can be easily integrated into workflow within a DAW using MIDI event sequencing. Nodal is a non-linear composition system in which artists are able to create generative systems consisting of cyclic structures and networks of events. Generative systems such as nodal are able to facilitate complex emergent structures, but this is often at the expense of the system's usefulness to create compositions in a traditional sense and requires users to think differently about the composition process. McCormack [8] addresses this paradigm as the focus of Nodal's design.

> The aim of the work...is to design and build a generative composition tool that exhibits complex emergent behaviour, but at the same time offers the composer the ability to structure and control processes in a compositional sense.

Nodal's design is intended to provide a fundamentally different approach to composition while attempting to retain the system's usefulness through intuitive visual mapping and simple event rules. Nodal (figure1) displays musical events as a grid of nodes; the space between nodes represents time and the connection between nodes indicates the direction of the next event. Nodes have the ability to be sequential, thats is they continue to follow a path or in the case of multiple nodes flow in sequential order. Parallel nodes take a single event and send it on multiple paths at once and random nodes select a path at random.

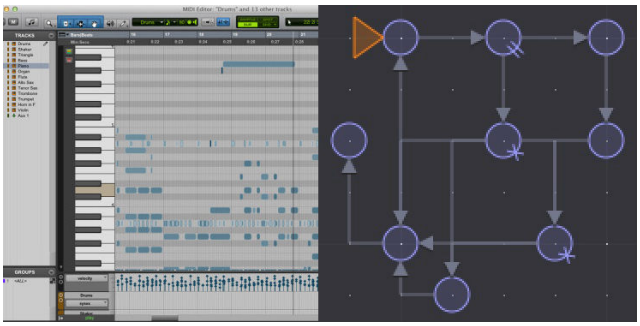


Figure 1: A Nodal network created by a participant

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2. DESIGN PARADIGMS FOR COMPOSITION

Digital composition systems are designed to allow musicians and music producers to deal with extremely complex amounts of musical information. These complex digital representations are inherently difficult to manage for users. Gaining a better understanding of how users develop strategies to complete complex compositional tasks in both traditional DAWs and novel composition systems will help to identify some of the fundamental differences in design that can be attributed to the usefulness of managing complexity within the system.

For users, learning and identifying tools and limitations afforded by a system is an iterative process and is simultaneous to the discovery of the potential compositional results the system is able to achieve [11]. This design problem for DAWs is summarised by Resnick et al. as "Low Threshold, High Ceiling, and Wide Walls". Effective tool designs should be simple enough for new users to get started, complex enough for experts to create sophisticated results and should support a wide range of explorations, such as different ways to control interaction with the system. [14] Additionally the opacity or the ability of the user to create a conceptual model of the system is paramount in enabling users to develop strategies easily. For new users opaque systems can lead to uncertainty, a lack of understanding and frustration.

Traditional DAWs are often focused on recording a performance of MIDI (Musical Instrument Digital Interface) or audio, in realtime. However after this performance has been captured the system's primary function is to edit and manipulate this information. Some component of this is the documentation of a performance for listening, but arguably the primary function of a DAW is to enable the composer to continue the composition process digitally. Ableton Live is a good example of this; the DAW has the ability to record audio, but the primary focus of it's design is the creation of electronic, or elctro-acoustic music. This is an example of a reversal of the traditional composition process: the composer creates the notation before the music is performed, and the musical structure may be sketched in abstract form before a more concrete instantiation is required [3].

One tool used frequently in composition systems to help users visualise sequences of events is the playhead. In Protools the playhead is represented by the scrolling bar which travels from left to right. This bar can be placed in linear time represented by the notated scale and playback will begin where it is placed. In Nodal the playhead is simply the change in colour that takes place when an event is triggered.

The systems used in this study require the user to develop different strategies to achieve complex musical compositions. Protools becomes more complex with the layering and intensity of sounds as well as increased frequency of events or variation within events, but Nodal becomes quite complex with few events as both randomness and chance can be introduced with a minimal series of events. In both cases feedback is provided to the user both aurally and visually. This feedback is the primary means by which the composer iteratively works with the system. Liveness [3] describes the type of feedback a user receives from a system as well as how responsive that feedback is when a user performs an action. Liveness as a measure is useful in defining the principal differences in feedback between composition systems.

Nodal incorporates some basic aspects of a music tracker such as VSTrack [12] as composition in nodal requires the mapping of at least some events prior to the execution of a sequence. However in contrast to a tracker the non-linear generative nature of Nodal makes it harder to follow when more complex event sequences take place. Nodal as well as being executable is edit triggered: while a sequence of events is playing edits to note sequence, pitch and timing are able to be made.

Protools is divided into three different interfaces track view, mixer, and midi editor. This study focuses on the piano roll found in the midi editor. The piano roll can be used to record performances and in that sense it is a stream-driven environment that is constantly active and continually visualising notations as they are made [3]. It can also be used to map events which can then be executed. This is limited to a single sequence of events: requiring any further sequences in the same pitch range to be divided amongst additional tracks.

Athanasopoulos et al. [1] have demonstrated in a cross-cultural study of people from the U.K., Japan and Papua New Guinea that in the absence of general or musical literacy people describe variation in time and pitch idiosyncratically. This highlights that some of the fundamental principals of western musical notation aren't necessarily self-evident. One of these being the concept of an independent sound object [4] which is inherent in western notation. For people literate in western standard notation a cartesian approach to musical notation is primarily adopted. But what was found when the BenaBena people of Papua New Guinea described musical pitch variation over time is that the concept of an abstract sound object was not evident. Structural variations that corresponded to pitch variation were seen as irrelevant and the participants attempted to illustrate the sound's mode of production through the use of culturally significant representations of the sound. This raises many questions for the design of novel composition systems. The usefulness of a system will always have to contend with user's cultural understanding of musical notation and the abstract concepts that are implied by that understanding. Mapping generative systems within a fundamentally cartesian understanding of musical notation dictates limitations. Determining if novel composition methods are better at facilitating the management of complexity is beyond the scope of this study, instead what is emphasised is the importance of the continued development of composition systems to include more than what is currently considered useful.

3. METHOD

This study aims to further understand how users develop strategies to complete complex tasks in linear and non-linear composition systems by looking at users experience and self-assessed ability to complete tasks. Primarily the study is focused on assessing three criteria:

1. Is the user effectively able to achieve complex compositional tasks?

2. What is the users experience of the system?

3. What are the relationships between the characteristics of the system and the musician's experiences and actions? [7]

Participants where asked to use both Nodal and Protools to achieve simple and complex compositional tasks. Participants were self described as having a musical background of varying types and skill levels. Prior to the commencement of the study a short demonstration was provided for each system to ensure a base line of knowledge for all participants. Tasks required users to develop and execute a

strategy for achieving the task and then to reflect on the system's design and their self-assessed level of ability to complete various components of the task. This was measured using two methods. Firstly a five point Likert scale from never to always with questions comparative to the Cognitive Dimensions of Notations framework [5] which focuses on comparing the user's reflections on the usability of computer based systems by identifying conceptual models employed by notational systems. It has similarly been applied to music typesetting as well as live-coding musical systems [2] Some questions included in the questionnaire were:

- I feel I am able to complete the task successfully
- I have total control over the system
- I have a strong sense of what I have to do to complete the task

After completing a task in both systems participants were asked to asses how they felt about the task as well their experience of each system comparatively. This was measured using a five point Likert scale from simple to frustrating. The questionnaire consisted of questions based on the psychometric scale used to measure flow. [13] Some questions included in the questionnaire were:

- Find the tools I need to use to complete operations
- Create new ideas I have spontaneously thought of
- Get the system to do what I want

Participants were also filmed during the study, this was to determine length of time they took to complete tasks, review strategies participants developed and iterations of compositions as they developed as well as record comments participants made about the process during the task. The outcome of these two methods of analysis and the video documentation is a detailed picture of a user's self-assessed experience and strategy when attempting to complete a task within a composition system.

The study consisted of seven participants of varying levels of musical ability and familiarity with DAWs. Each participant was asked to complete two tasks in both Protools and Nodal. The first task was to create a soundscape for a video game. The soundscape had to be sufficiently long enough that it would not become boring or repetitive when used in the video game, requiring participants to create variations on a theme. The second task was to create an improvised solo to the twelve bar blues. The twelve bar blues was provided for the participants, and they were allowed to copy and paste from the existing patterns to create new ones. This task required participants to pay careful attention to melodic structure, harmony, disharmony and timing.

4. RESULTS

4.1 Reorganisation Of Events

The study identified that regardless of the participants level of experience with composition systems, they found reorganising sounds in order to modify the order of something they had composed harder in Nodal than in Protools. (Figure 2)

For example, participants with more than one year experience using DAW's (Protools and Ableton) did not find it any easier than participants with less than one year or no prior experience using DAWs. This result identifies an inherent design in non-linear systems. Manipulation of events within linear timeline based systems involves direct manipulation of events, such as the alignment and placement of notes on a piano roll. Whereas generative systems are designed to allow users to set up systems of events which then take place over time, which is the indirect manipulation of events. One implication of this is deciphering the outcomes of notations, in order to further understand how to achieve a desired result, is a slow and difficult process. This is supported by the study's results. When participants were asked if they felt they have total control over the system when completing a task, 57% answered frequently in protools, compared to 28% in Nodal. Once a Nodal Network becomes sufficiently complex it becomes difficult to visually track events. This lowers the amount of tangible feedback provided by the system.

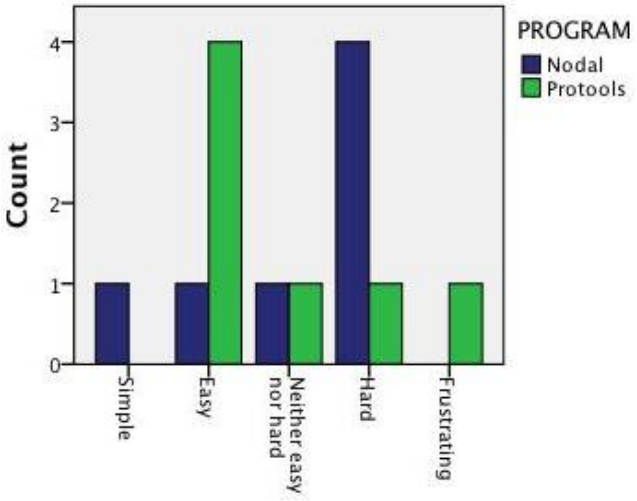


Figure 2: Reorganise sounds and modify the order of something I have made

4.2 Error Correction

The study additionally identified that regardless of aparticipants level of experience using DAWs, they found correcting mistakes in Protools easier than in Nodal.

Indirect manipulation of events within non-linear systems often adds to opacity. In Nodal it is most evident when modifying events during playback. The ability of Nodal to create complex sequences of events based on simple rules is intuitive when composing a network, but once playback has begun the sequence becomes difficult to follow, particularly when multiple events are taking place at once. Visualisation of the playhead, the change in colour of a node when triggered is the primary form of visual feed back. This is hard to follow when the system becomes sufficiently complex. Protools timeline and piano roll also contains the potential to become sufficiently complex to make the visual tracking of events difficult. But coupled with the ability to stop and start playback from a selected point within the timeline allows for quick and iterative changes to compositions.

4.3 Ease Of Use

Athanasopoulos et al. [1] study showed that individuals familiar with reading left to right and modern western musical notation were more likely to represent musical information in that way. This potentially means that familiarity of participants with the piano, and the left to right linear playhead in Protools means participants are more likely to intuitively understand Protools UI. This is reflected in the user study's result to the question "I do things easily without thinking".

71.4% of participants using Protools answered frequently compared to 28.57% when using Nodal. Users with little or no experience with DAWs will still have some level of understanding and a cultural knowledge that might aid them in interfacing with the system. Nodal's less familiar mapping requires more of the user, this is expressed by Murray et al. [11] who show that users are simultaneously learning and discovering the potential compositional results attainable with the system. While the results of this study are not statistically significant because of the small number of participants, this result also points to the potential that the piano roll is in fact an extremely intuitive way to represent the division of pitch. Sequential division of frequencies in a 2D plane could be well represented by the piano roll.

4.4 User Experience

Throughout the study participants were told they could ask questions and converse about what they are doing. During these conversations some common themes emerged. The primary theme that developed was that even though users feel less confident in their ability, feel as though they have less control and have to think harder about what they would like to do, they find composition in Nodal more rewarding. Additionally people using Nodal seem to be more absorbed in the task they are completing. This is evident in the response to the statement "Time seemed to pass quickly", 85.7% of participants using Nodal responded to the question with Always, comparative to 42.9% when using Protools.

Generative systems can often take on a game-like design. There are many examples of video games which incorporate generative systems into there design such as Iwai's Otocky[10]. User experience and usability create different criteria for user interface design. Games such as Otocky require more effort and are extremely limiting for creating compositions but can conversely result in a more enjoyable and fun experience for users [6]. Nodal's novel approach to composition is somewhat comparable, particularly in terms of users' conscious and unconscious control of events. Users retain less immediate control of events but find composition more enjoyable. This is supported by the response to the statement "using the system is extremely rewarding" where 71.4% of users responded always. Additionally, of the participants who self identified themselves as inexperienced with generative and digital composition systems but as experienced musicians, most had trouble initially understanding the concepts of a generative system and how it could be legitimately useful. But after using the system once or twice they remarked on how they could see how Nodal's approach to sequencing events was interesting, several participants expressed there excitement in the creative potential Nodal has. However the participants who did express their interest in Nodal did not necessarily create more complex or musically interesting results.

4.5 Strategies In Nodal

Participants who were filmed adopted several strategies to solve the composition tasks they were given. Strategies were consistently developed by participants as the tasks went on. When participants began the second task (improvisation to the twelve bar blues) in each system they usually took on a similar style of strategy as they did in the first task (compose a sound scape for a video-game). One result was that users who were familiar with linear DAW's seemed to understand how to create more musically complex results in both systems, even when they were less experienced musicians. The two figures (3 and 4) show the results of two participants who demonstrated this finding. Figure 3. is a system composed by a participant who self-assessed as a non-musician but who had experience using popular DAWs such as protools. The participant begun with a simple set of events and created a loop, with a melodic sequence. Then they added a second instrument and then added in the parallel nodes. The result was an exponential gain of events, this is as parallel events were triggered more events were created and then triggered again and so on. An infinite loop of events triggering at all times was the outcome. To solve this the participant begun to explore the idea of a dead end, or a drain for the excess events. It took several iterations of drains until the system was sufficiently consistent with creation of events and draining of events. The participant has effectively created a cyclic composition that creates variations in six note cycles. The network incorporates a drain which removes the additional events created by the parallel nodes and a second instrument which triggers at a separate point in the network to the first. This demonstrates an in-depth understanding of Nodal and the potential it has to great complex variation. This system illustrates how simply and efficiently Nodal can become complex. Figure 4. is a system composed by a self-assessed proficient musician. The participant has developed an understanding of rhythm and repetition in Nodal but has not developed the network beyond a linear sequence of events. This result became consistent across the participants who where inexperienced with DAWs but were experienced musicians. While the result is not statistically significant it suggests that as trained musicians have developed an advanced and habitual composition process, they may find incorporating a simple and less controlled methodology into their composition process a challenge at first. Another possible reason for this result could be that Nodal has some core concepts built into it that are commonly found in most digital composition systems. MIDI is considered a standard communication protocol across the vast majority of digital composition systems.

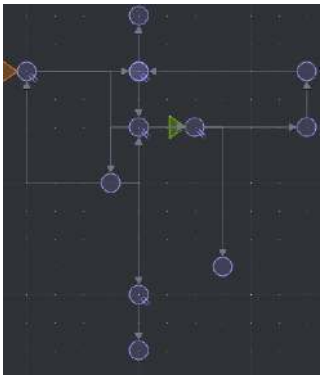


Figure 3: complex system produced by an experienced participant with little musical background

5. DISCUSSION

Nodal has shown to be effective in facilitating complex compositions with simple tools which are easy to understand and use. This has been achieved through the use of intuitive mapping of events and simple structural concepts. This study identifies the potential usefulness of non-linear and generative systems such as Nodal to facilitate a better user experience for musicians when composing. While

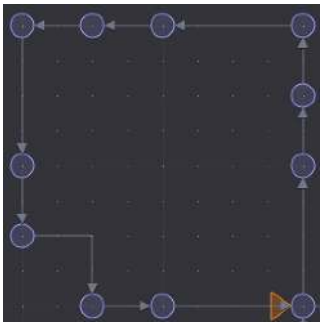


Figure 4: Simple system, produced by a user with little previous DAW experience but with significant musical experience

DAWs such as Protools are clear and precise tools for creating compositions they lack the compelling ability to organise events as non-linear networks of events. Protools piano roll and linear track view are both more immediate and intuitive but less interesting to new users and often less creative to inexperienced musicians. To create more effective tools that incorporate non-linear deign the identification of potential design problems is paramount. Participants in this study have identified several key design problems that can be broadly applied to non-linear composition systems.

- The ability to track events through the use of the representation of the playhead is key in user's ability to correct errors and make simple changes to the structure of events.
- Users potentially find non-linear composition tools more rewarding, but they are not necessarily as useful in creating linear compositions.
- The potential of non-linear systems is not immediately obvious
- Experience with generative systems does not necessarily make it easier to understand Nodal as a first time user.
- Indirect manipulation of events makes visualising the sequence of events harder once the system is sufficiently complex.

One key design factor that has become clear in this study is that the design of composition systems which are only linear or non-linear is limiting. Incorporation of both elements is what could potentially create powerful tools for flexible and complex creativity within a composition. Nodal and Protools both integrate MIDI as an input and output method for communicating with other systems, as well as using terminology and functionality internally that is specific to MIDI functionality. In Nodal you have the ability to define pitch by reference to the note and octave relative to a piano (C6,D4 etc...), this is identical in principal to the piano roll in Protools. Both have inherent design affordances, being able to map sequential events between both paradigms could allow composers to maintain flexible yet fine control over the composition. Generative systems are unique systems which when set in motion contain some degree of autonomy contributing to the completed work of art. Designing digital composition systems which allow users to create autonomously and are at the same time able to allow for fine control of events could enable composers to use generative processes without sacrificing control.

There is a tension between usability and user experience in the design of composition systems. The goal of composition systems is to both facilitate complex and creative tasks. In fulfilment of these goals a composition system should be effective and efficient at completing tasks, easy to learn and to remember. But at the same time it must also be motivating, helpful, rewarding and entertaining so as to be conducive to creative outcomes.

6. FURTHER STUDY

Study into similar composition systems will allow for the furthered verification of these design problems. Beyond that, interviewing experienced electronic artists who are using generative processes in their compositions could potentially identify additional factors which musicians experience when using these systems. Understanding the potential benefits of experimental composition systems will broaden the scope of design of tools for composers, aiding the creative process and creating better end user experience.

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